Homework: English for Mathematics 1
Autumn Term 2022 – MA33002EHIT

Name:_____王嘉禾_____ Student ID:_____2022110615______

Submission deadline is **September 23rd,18:00**.
Submission:

- by e-mail to f.bossmann@hit.edu.cn
  Use your university e-mail address (…@stud.hit.edu.cn).
  Use "Homework [your student number] [your course number]" as subject.
  Attach the homework as **one pdf file**. Do not use any cloud service.
- In the last class (September 20th)
  Place the printed homework on the teacher's desk until end of class.
- At my office
  Bring the printed homework to my office on September 23rd, 13:00 – 18:00.
  My office is in Science Park, TIB building, room K1344.

Homework rules:

- The homework consists of five exercises, each giving 20 points (a total of 100 points). You need 60 points to pass.
- Use **American English** if not stated otherwise.
- You can use the lecture notes for help. However, **do not copy the examples from the lecture notes** word by word. You need to come up with your own answers. Any direct copy from the lecture notes will not count as an answer.
- Every student needs to fill out his own homework. Do not copy your homework from another classmate. **If I find homework that was copied from someone else, both submissions will be invalid.**

Exercise 1: (Chapter 3, 1 point each)

Read the following examples carefully. Each example violates one of the 11 rules of Chapter 3. Write the number of the rule that is violated in the box on the right (See the first line as example).

Hint: Each rule has its exceptions.

The function f is: convex, bounded, and differentiable.	7
The function f is convex but the function g can be non-convex.	4
We do not know of any method that can solve this problem, the non-convex function f causes too many difficulties.	
This result is used in publication [3], [4] and [5].	2
This technique – also know as gradient descent – is quite popular.	8
The sets S_1, S_2, and S_3 all contains the point x=0.	9
This algorithm converges linear. Which will be shown in later sections.	6
Note that Theorem 10 in contrast to Theorem 8, also applies to quadratic functions.	3
There exists no function that is convex concave, and non-linear.	2
Being differentiable, we can calculate the minimum of this function.	11
She published this work by her.	10
We revised our work to implement the reviewers comments.	1
We can choose x greater 0 and apply Newtons method.	4
The reviewer gave useful suggestions, we submitted a new version of our work.	5
We discuss three different topics on this algorithm accuracy, efficiency, and possible applications.	7
There are no positive integer smaller than 0.	9
None of theirs works cover this topic.	10
The variable is positive i.e. it is greater than 0.	3
Charles' lemma applies to functions that are: convex and linear.	7
Being an expert on this topic, his works are most important.	11
The parameter choice is complicated. Because of the unknown setup.	6

Exercise 2: (Chapter 4, 3+3+3+5+3+3 points)

Remember: Do not copy any of the examples form the lecture notes!

a) Why should you use the active voice in your text?
   Give one example where the use of the passive voice is acceptable.
   State why it is acceptable in your example.
   • Because the active voice is often,shorter,easier to read,and more clear in its meaning than the passive voice. It also can sound much stronger.
   • This method is first proved by Newton in 1666
   • In some cases , the active form and passive form is slightly different,so the right choice depends on the context .For the upward sentence,it may be more suitable for "The method"we mentioned,has just been shown or just has been mentioned in the sentence before.

b) Give an example sentence containing a negative form.
   State why it is wrong or right to use the negative form in your example.
   • We will not prove this theorem in our work
   • That is wrong. Because the reader may think, why? why this theorem needn't to be proved.Thus they may not focusing on our work, or begin to doubt the authority of our work.

c) Sometimes explaining a concept or idea can be complicated using only plain written text. What other things can we use in scientific writing to help the reader understand our work? (list **three** things)
   • Definite language
   • Specific language
   • Concrete language

d) Each word in a sentence should give some contribution. This can be by
   - giving relevant information,
   - emphasizing parts of the sentence or changing the connotation,
   - or just being necessary to form sound a correct sentence.

   Any word not belonging to one of the three above categories can be considered needless. Give an **example sentence with at least 10 words** that contains at least one word of each category and at least one needless word. Mark all words according to their category.
   • Surprisingly, we can use the fact that every nonempty set of real numbers which is bounded from above has a unique least upper bound to prove the existness and the uniqueness of the set X's upper bound.
   • necessary information emphasis needless

e) Give an example text that explains two similar ideas, concepts, … .

- Why i can see the absolutely same answer in Petter's work, Jack's work, and your work?

f) Give an example for Rule 22. Underline the two parts of your sentence that contain the most important information.

- Except my method, there are no method left can prove the function f is bounded from above.

Exercise 3: (Chapter 6, 2 points each)

Fill in the missing words. Use the words and expressions of Chapter 6.

Comparing publication [3] ____to____ publication [4], we see that both works use similar ideas.
We use the set that was defined in the last section and the function f which is given in equation (3).
This homework is ____divided into____ several exercises.
There is no suitable choice ____among____ the set of linear functions.
Compared ____with____ newer techniques, the old method performs only slightly worse.
This argument is the same _which_ was given in the last chapter.
When the input is __less__ noisy, we need ____fewer____ iterations.
The set can be ____divided into____ several subsets.
We have to balance the method ____between____ numerical stability and efficiency.
Figure 3 shows the data that ____is____ gathered in these experiments.

Exercise 4: (Chapter 8, 2.5 points per category)

Chapter 8 provides linking words divided into 8 different categories. Name each category and give **two examples for each** (using a **different linking word** in each example). Do not copy any of the examples from the lecture notes!

1. Sequence)
   - First: First, let's talked about the case if function f is convex
   - In conclusion: In conclusion, we can see the function f is convex, bounded, and linear
2. Addition)

• And: The function f is convex and bounded.

• Furthermore:The function is not only convex, furthermore,it is bounded.

3. Cause)

• Since: Since the nonempty set X is bounded, it has the least upper bound

• Due to: There must exists a “c”, due to The Complete Axiom.

4. Effect)

• So: The set X is bounded, so it has the least upper bound.

• As a result: The set X is empty, as a result, the assumption is false.

5. Emphasis)

• Indeed: To prove the theorem, we indeed need to build a set X, that is nonempty.

• Obviously: Obviously, The theorem is wrong.

6. Comparison)

• Similarly: This method can prove that theorem10 is right, similarly, it can prove theorem11 is right.

• Also: The theorem is holds on R, also on C.

7. Example)

• Such as:The function f has many properties, such as convex, bounded, and linear.

• For example: We can list some important fields, for example, N ,Q ,R ,andC.

8. Contrast)

• However: The function f has a maximum, however ,it doesn’t has a minimum.

• But: We have listed a lot of condition before, but, they are still not enough to prove this theory.

Exercise 5: (general questions, 4 points each)

a) Name three different building blocks of text.
Which building block should be the smallest containing self-contained unit?
- Text Chapter Section
- Section

b) What is additional and required information?
Give an example for both and underline the information.
- Additional information means the meaning of the sentence doesn't change when we delete them, they just explain the former or add some information.While required information means the meaning of the sentence will change when we delete them.
- The natural number field N, which contains 1,2,3,4,and so on, is an important field.

Additional: "which contains 1,2,3,4,and so on"
Required: "natrual number field N"

c) Write a sentence once in American English and a second time in British English. Choose your sentence such that **at least three words are different**.
- American English: The data we collect today,is different than yesterday, e.g.,the maximum of today is 500 but yesterday was100.
- British English: The data we collect today, are different from yesterday, e.g. the maximum of today is 500 but yesterday was100.

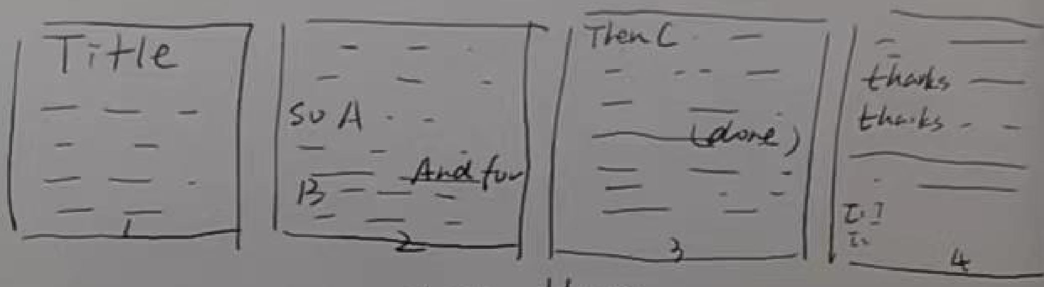
d) Text is mostly linear, i.e., it is read from the beginning to the end. Why can this be a problem in scientific writing?
- Because the reader may read fast or impatiently, and they may skip some words.

e) Text is an encoding of information or ideas we have in our mind. What is the problem of "information flow" in this context? (You can draw a sketch.)
On the next page.

note: If you have there important opinions in the passage.
A.B.C

Then you put them in the passage like A:

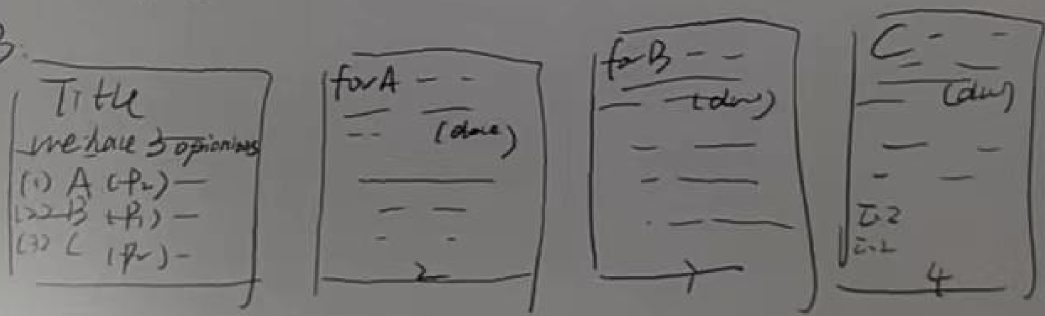
A.



The reader may skim them.

But if you put them like B:

B.



The reader may catch them ~~easily~~ easier.